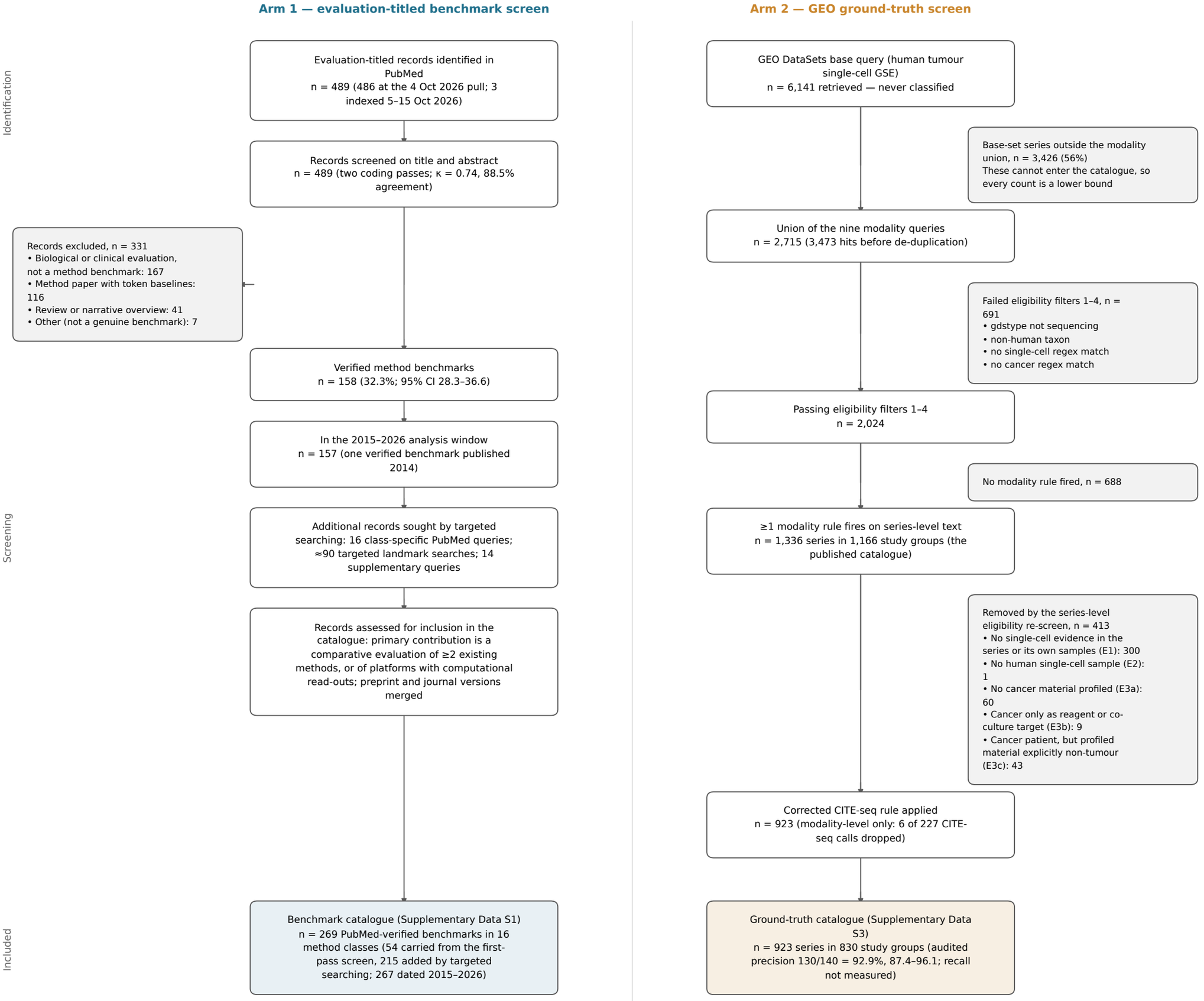
Arm 1 — evaluation-titled benchmark screen
Arm 2 — GEO ground-truth screen
Identification
Screening
Included
Evaluation-titled records identified in PubMed
n = 489 (486 at the 4 Oct 2026 pull; 3 indexed 5–15 Oct 2026)
Records screened on title and abstract
n = 489 (two coding passes; κ = 0.74, 88.5% agreement)
Records excluded, n = 331
• Biological or clinical evaluation, not a method benchmark: 167
• Method paper with token baselines: 116
• Review or narrative overview: 41
• Other (not a genuine benchmark): 7
Verified method benchmarks
n = 158 (32.3%; 95% CI 28.3–36.6)
In the 2015–2026 analysis window
n = 157 (one verified benchmark published 2014)
Additional records sought by targeted searching: 16 class-specific PubMed queries; ≈90 targeted landmark searches; 14 supplementary queries
Records assessed for inclusion in the catalogue: primary contribution is a comparative evaluation of ≥2 existing methods, or of platforms with computational read-outs; preprint and journal versions merged
Benchmark catalogue (Supplementary Data S1)
n = 269 PubMed-verified benchmarks in 16 method classes (54 carried from the first-pass screen, 215 added by targeted searching; 267 dated 2015–2026)
GEO DataSets base query (human tumour single-cell GSE)
n = 6,141 retrieved — never classified
Base-set series outside the modality union, n = 3,426 (56%)
These cannot enter the catalogue, so every count is a lower bound
Union of the nine modality queries
n = 2,715 (3,473 hits before de-duplication)
Failed eligibility filters 1–4, n = 691
• gdstype not sequencing
• non-human taxon
• no single-cell regex match
• no cancer regex match
Passing eligibility filters 1–4
n = 2,024
No modality rule fired, n = 688
≥1 modality rule fires on series-level text
n = 1,336 series in 1,166 study groups (the published catalogue)
Removed by the series-level eligibility re-screen, n = 413
• No single-cell evidence in the series or its own samples (E1): 300
• No human single-cell sample (E2): 1
• No cancer material profiled (E3a): 60
• Cancer only as reagent or co-culture target (E3b): 9
• Cancer patient, but profiled material explicitly non-tumour (E3c): 43
Corrected CITE-seq rule applied
n = 923 (modality-level only: 6 of 227 CITE-seq calls dropped)
Ground-truth catalogue (Supplementary Data S3)
n = 923 series in 830 study groups (audited precision 130/140 = 92.9%, 87.4–96.1; recall not measured)